

Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
		(ii)		enantiomers (1) neutral answer – stereoisomers racemic mixture / racemate (1)	2					2
	(c)			(stereoisomerism occurs when two forms of a compound have) the same structure but different positions in space / different 3D arrangement	1					1
	(d)			$\text{HOOC}-\underset{\text{H}}{\text{C}}=\underset{\text{H}}{\text{C}}-\text{COOH} + \text{H}_2\text{O} + [\text{O}] \rightarrow \text{HOOC}-\underset{\text{OH}}{\text{C}}(\text{H})-\underset{\text{OH}}{\text{C}}(\text{H})-\text{COOH}$			1			1
	(e)			polar OH / COOH groups dissolve by hydrogen bonding (1) but as the hydrocarbon chain length increases the polar groups form an increasingly smaller part of a largely non-polar molecule (1)	1	1				2
				Total question 8	7	3	3	1	1	13